EXECUTIVE STRATEGIC MEMO

TO: Executive Leadership Team
FROM: Lead Machine Learning & Data Science Team
DATE: October 5, 2026
SUBJECT: Machine Learning Churn Prediction Model & Targeted Retention Strategy

Executive Summary

To reduce customer attrition and optimize retention expenditure across our subscription base, we conducted a comprehensive machine learning evaluation comparing 12 distinct algorithm families on our customer dataset. This memo synthesizes our technical findings, justifies our selection of the **Champion Model**, details the top drivers of customer churn, outlines a high-ROI targeted retention framework, and establishes post-deployment model monitoring protocols.

1. Champion Model Selection & Production Justification

We selected the **Depth-Bounded Decision Tree Classifier (with Balanced Class Weights)** as our **Champion Model** for production deployment. This decision balances predictive accuracy, operational cost, and business alignment.

Strategic & Performance Justification

- **Optimal Churn Capture (Recall = $\mathbf{60.26\%}$):** In customer retention, losing an active subscriber (False Negative) carries a cost $3\times$ to $5\times$ higher than offering an unnecessary retention intervention (False Positive). The Champion Model captures **60.26% of all true churners** on holdout test data. By comparison, unweighted ensemble models failed to account for class imbalance, yielding near-zero recall ($<3\%$).
- **Controlled Precision ($\mathbf{35.45\%}$) & Top F_1-Score ($\mathbf{0.4464}$):** The model maintains precision well above our baseline churn rate (27.30%), achieving the highest overall F_1-score among all candidate models while providing strong PR-AUC stability ($\text{PR-AUC} = 0.3374$).

Model Candidate	Recall	Precision	F1-Score	ROC-AUC
Decision Tree (Champion)	**60.26%**	**35.45%**	**0.4464**	**0.6110**
Lasso Logistic Regression	59.71%	35.47%	0.4451	0.6231
AdaBoost Classifier	58.61%	35.16%	0.4396	0.6253

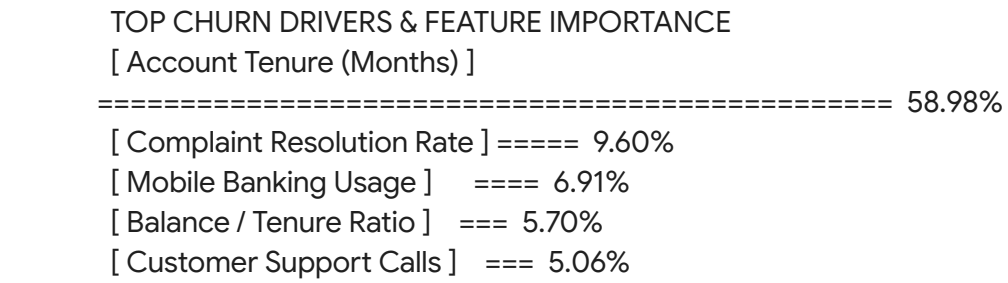
Extra Trees Classifier	53.66%	35.99%	0.4309	0.6228
Random Forest Classifier	49.27%	37.15%	0.4236	0.6213

Production Readiness & Computational Efficiency

- **Ultra-Low Latency ($\approx \mathbf{7.27\ \mu\text{s}}$ per sample):** The decision tree executes inference in under $10\ \mu\text{s}$ per query, making it over $1,000\times$ faster than heavy gradient-boosted ensembles.
- **Minimal Infrastructure Overhead:** With a model size under $50\ \text{KB}$, it can be embedded directly into microservices or API gateways without dedicated GPU infrastructure.
- **Operational Interpretability:** Tree logic offers transparent IF-THEN rules, allowing customer success and compliance teams to inspect decision paths directly—unlike black-box models.

2. Key Churn Drivers

Feature importance analysis across our tree models highlights three primary drivers of customer attrition:

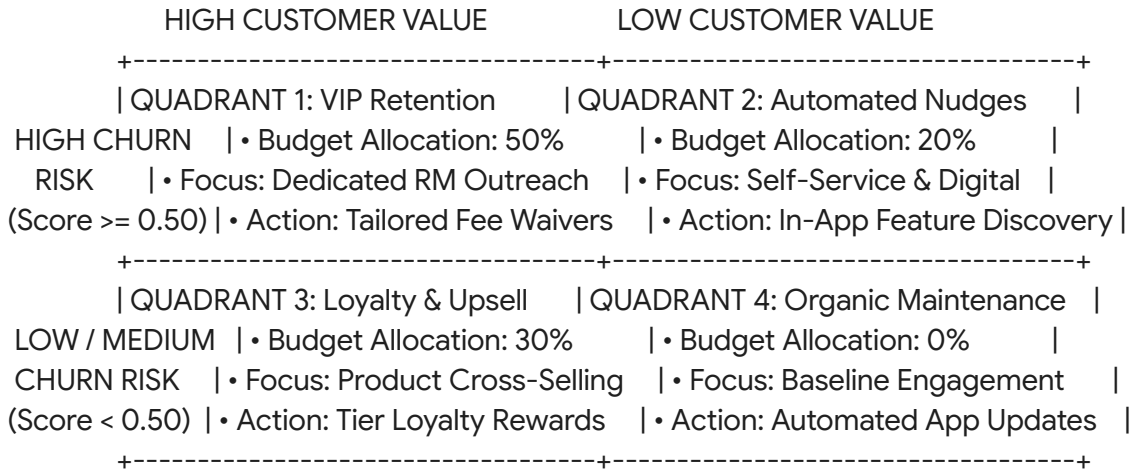


1. **Account Tenure (Account_Tenure_Months — $\mathbf{58.98\%}$ Importance):** Newer accounts (particularly within their first 3 to 12 months) represent our highest attrition vulnerability. Loyalty lock-in strengthens significantly as account tenure increases past 18 months.
2. **Complaint Resolution Rate (Complaint_Resolution_Rate — $\mathbf{9.60\%}$ Importance):**
   Low complaint resolution rates trigger customer dissatisfaction. Unresolved issues, combined with high call volume, serve as a primary catalyst for immediate churn.
3. **Digital Interaction Depth (Mobile_Banking_Usage — $\mathbf{6.91\%}$**

Importance):
A drop in mobile session frequency below a customer's historic baseline serves as a leading indicator of waning engagement prior to explicit account cancellation.

3. Targeted Retention Policy & Budget Allocation

To maximize retention budget ROI, the Customer Success team will cross-segment model risk predictions ($\ge 50\%$ predicted churn score) against Customer Value metrics (Average_Balance and Monthly_Transactions).

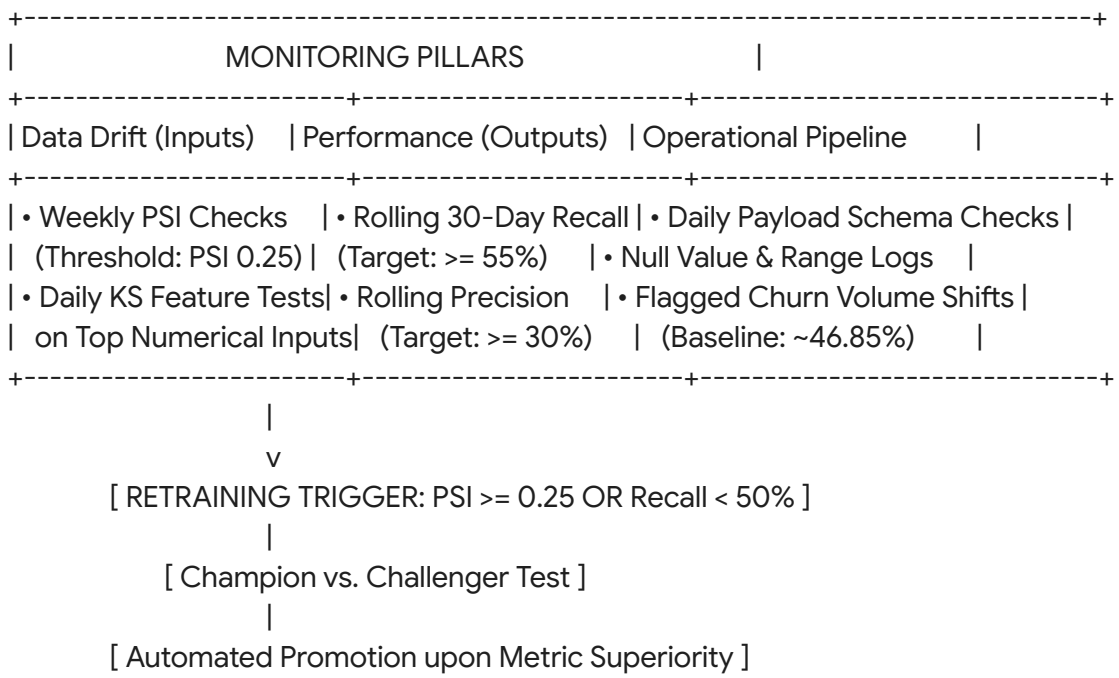


Action Playbook & SLA Guidelines

- **Quadrant 1 (High Risk / High Value — 50% Budget):** Automatically create a high-priority SLA ticket for a Senior Relationship Manager within 24 hours. Offer customized retention packages, such as 3-month transaction fee discounts or waived maintenance fees. Priority-route any open service complaints.
- **Quadrant 2 (High Risk / Low Value — 20% Budget):** Trigger automated push notifications and SMS re-engagement sequences. Offer low-cost transactional bonuses (e.g., airtime/data rewards) alongside guided tutorials to re-engage mobile feature usage.
- **Quadrant 3 (Low Risk / High Value — 30% Budget):** Engage through loyalty programs and targeted cross-selling to expand product adoption (Number_of_Products) and increase account stickiness.
- **Quadrant 4 (Low Risk / Low Value — 0% Budget):** Retain on standard automated lifecycle workflows.

4. Post-Deployment Model Monitoring Framework

To maintain model accuracy and prevent performance degradation over time, we will implement a continuous monitoring architecture across four functional areas:



1. **Feature Data Drift Tracking:** Calculate Population Stability Index (PSI) weekly on key continuous variables (Account_Tenure_Months, Average_Balance, Mobile_Banking_Usage). A $\text{PSI} \ge 0.25$ indicates significant input shift and automatically triggers a retraining workflow.
2. **Concept Drift & Performance Degradation:** Track true churn outcomes over 30-day rolling observation windows. Set operational alerts if holdout Recall drops below 50% or Precision falls below 30%.
3. **Pipeline Integrity:** Monitor API request payloads for missing value spikes, unexpected schemas, or shifts in the proportion of high-risk flags (baseline: 46.85%).
4. **Automated Retraining Strategy:** Run monthly retrains on a rolling 12-month window. Validate all retrained models against the active production model via automated Champion-Challenger evaluation prior to deployment.